CHEAT SHEET UTS · WEEKS 1–7

MK 2 · Financial Management & Strategy

Don't mix these up

- Profit ≠ cash. Assets up = cash out; liabilities up = cash in.
- Cash flow: buying fixed assets = investing; dividends & debt = financing; operating = core business.
- Product costs → inventory → COGS when sold. Period costs → expensed. Ignore sunk costs.
- Production > sales → absorption profit > variable (fixed overhead parked in inventory).
- Higher discount rate → lower PV. Rates up → bond prices down. Compare loans by EAR.
- Annuity due = ordinary annuity × (1 + r). Growing perpetuity needs r > g.

W1 Foundations of Financial Management & Accounting

The bridge: decisions → records → reports → decisions · What is finance, and what is the firm's goal? · The financial manager's three decisions and guiding principles · Accounting: the reporting language of business · Four statements as one story · How the numbers connect (MiniMart) · The 9–step accounting cycle · The multi-step income statement in one chart · From opening to closing cash: the MiniMart story

- Goal: long-term shareholder wealth, not short-term profit.
- Profit misses timing, cash flow and risk.
- Three decisions: investment, financing, working capital — all on the balance sheet.
- Assets = Liabilities + Equity; every transaction keeps balance.
- Four statements are four views of one story; chase the same number across them.
- Profit ≠ cash; operating cash flow is key to sustainability.
- Cycle: analyze → journalize → post → trial balance → adjust → report → close.

FORMULAS

```
ASSETS = LIABILITIES + EQUITY
```

```
Laba bersih ± item non-kas ± perubahan modal kerja = Arus kas operasi
```

W2 Financial Statement & Ratio Analysis

Reporting vs analysis · Earnings quality: profit isn't always performance · 6–step framework and tools · Five ratio families · DuPont analysis: decomposing ROE · Ratios in forecasting (bridge to Week 3) · The four financial statements as one system · From ratio signal to action, and how to compare · Bartlett's ratios charted · Try it: the DuPont calculator

- Reporting = information; analysis = judgment for capital allocation.
- The notes are often where analysis really begins.
- Computing isn't analyzing — ratios need context.
- Vertical/horizontal common-size; watch receivables/inventory outgrowing sales.
- Five ratio families: liquidity, activity, solvency, profitability, valuation.
- DuPont: ROE = margin × turnover × leverage; leverage-driven ROE is fragile.
- Forecasting uses historical ratios plus strategy, industry and macro judgment.

FORMULAS

```
Current ratio = Aset lancar ÷ Liabilitas lancar
```

```
Quick ratio = (Kas + Surat berharga + Piutang) ÷ Liabilitas lancar
```

```
Debt ratio = Total liabilitas ÷ Total aset
```

```
Debt-to-equity = Total utang ÷ Total ekuitas
```

```
Interest coverage = EBIT ÷ Beban bunga
```

```
ROE = Net margin × Asset turnover × Financial leverage
```

W3 Cash Flow & Financial Planning (Cash Budgets & Pro Forma)

Depreciation: an expense that uses no cash · Measuring cash flow: sources vs uses, OCF, FCF · Cash budget: forecasting shortfalls and surpluses · Pro forma statements: projected income and balance sheet · Why is Baker's CFO $837 but OCF $392.3? Two 'operating cash flow' numbers · Cash flow within the month and the personal budget · Why faster depreciation means more cash early · Coulson's cash budget charted: profit ≠ cash · Percent-of-sales vs fixed/variable split: a big difference

- Depreciation is a noncash charge; faster depreciation = lower early taxes.
- Asset increase = cash out; liability increase = cash in.
- OCF = EBIT(1–T) + Dep; FCF = OCF – NFAI – NCAI.
- Cash budgets reveal borrowing needs in advance — arrange credit lines early.
- Use scenarios — worst-case needs can far exceed the base case.
- Percent-of-sales ignores fixed costs; split fixed vs variable.
- External financing required is the balance-sheet plug.

FORMULAS

```
OCF = EBIT(1-T) + Dep = 392.3
```

```
FCF = OCF - NFAI - NCAI = 392.3 - 300 - 57 = 35.3
```

```
OCF = NOPAT + Depreciation = EBIT × (1 - T) + Depreciation = 370 × 0.79 + 100 = 392.3
```

```
Year 2: $12,800 × 21% = $2,688
```

W4 Managerial Accounting & Costing Systems (Job-Order & Process Costing)

Cost classification depends on purpose · Cost behavior and decision costs · Job-order costing: for unique orders · Process costing: for uniform mass production · Product vs period costs, and the cost of goods manufactured (COGM) schedule · Job-order journal flow and under/overapplied overhead · High-low drawn: separating fixed and variable cost · Job KR-118 cost and the job-order cost flow · Equivalent units: why half-finished goods count partly

- Cost classification depends on purpose.
- Product costs → inventory → COGS; period costs expensed immediately.
- Mixed cost Y = a + bX; separate via high-low or regression.
- Ignore sunk costs; opportunity costs are relevant.
- Job-order: cost per job; overhead applied via POHR.
- Process: cost per department; equivalent units; reconcile.
- Average unit cost ≠ incremental cost.

FORMULAS

```
Prime = DM + DL
```

```
Conversion = DL + MOH
```

```
Biaya campuran: Y = a + bX (a = biaya tetap, b = biaya variabel per unit aktivitas, X = aktivitas)
```

```
POHR = Estimated overhead ÷ Estimated allocation base; Applied = POHR × actual base
```

```
Unit ekuivalen = Unit selesai & ditransfer + Unit ekuivalen di WIP akhir
```

```
Biaya per unit ekuivalen = (Biaya WIP awal + Biaya ditambahkan) ÷ Unit ekuivalen
```

```
COGM = DM used + DL + MOH applied + Beginning WIP - Ending WIPCOGS = Beginning finished goods + COGM - Ending finished goods
```

```
Transferred 142.8m + ending WIP 3.3m = 146.1m ✔
```

W5 Cost-Volume-Profit, Variable Costing & Activity-Based Costing

CVP: five moving parts and contribution margin · Break-even, target profit, margin of safety, operating leverage · Variable vs absorption costing, and segment reporting · Activity-Based Costing (ABC) · Model limits: CVP assumptions, customer profitability and ABC weaknesses · See it: the CVP graph and a break-even calculator · Why variable and absorption costing profits differ · ABC in one picture: where overhead really comes from

- Contribution margin covers fixed costs, then becomes profit.
- Break-even = Fixed ÷ unit CM (or ÷ CM ratio); target units = (Target + Fixed) ÷ unit CM.
- Margin of safety = sales cushion before loss.
- DOL = CM ÷ operating income; high fixed costs amplify swings.
- A shift to low-CM products raises break-even.
- Absorption vs variable costing differ only in fixed overhead.
- ABC: resources → activities → cost objects; often reveals loss-making custom products/customers.

FORMULAS

```
CM = Sales - Variable costs
```

```
Break-even (unit) = Biaya tetap ÷ Unit CM  |  Break-even ($) = Biaya tetap ÷ CM ratio
```

```
Unit untuk target laba = (Target laba + Biaya tetap) ÷ Unit CM
```

```
MOS = Penjualan aktual - Penjualan impas; MOS% = MOS ÷ Penjualan aktual
```

```
DOL = Margin kontribusi ÷ Laba operasi; %Δ Laba = DOL × %Δ Penjualan
```

```
Segment margin = CM segmen - Biaya tetap traceable
```

```
Profit difference = FMOH per unit × change in inventory units
```

W6 Time Value of Money I — Single Amounts & Annuity Basics

Why timing matters: compounding vs discounting · Future and present value of a single amount · Annuities, annuities due and perpetuities · Key ideas behind the formulas · Worked numbers: ordinary annuity vs annuity due, and mixed streams · Seeing compounding: the FV and PV curves · Garuda vs Samudra: comparing properly

- Money now > money later; compare at one point in time.
- $FV = PV(1+r)^n$; $PV = FV \div (1+r)^n$.
- Discount rate = opportunity cost, reflecting project risk.
- Annuity due > ordinary annuity (× (1+r)).
- Perpetuity: CF ÷ r; growing: CF ÷ (r – g).
- Compare rates using EAR, not nominal rates.
- Largest nominal total ≠ largest PV; delay destroys value.

FORMULAS

```
FV = PV × (1 + r)n; PV = FV ÷ (1 + r)n
```

```
FV anuitas biasa = CF × [(1 + r)n - 1] ÷ r
```

```
PV anuitas biasa = CF × [1 - (1 + r) -n] ÷ r
```

```
Anuitas di muka = nilai anuitas biasa × (1 + r)
```

```
PV perpetuitas = CF1 ÷ r  |  PV perpetuitas tumbuh = CF1 ÷ (r - g)
```

```
PV of ordinary annuity = CF × [1 - 1/(1+r)ⁿ] ÷ rPV of annuity due = PV of ordinary annuity × (1 + r)
```

W7 TVM II — Effective Rates, Loans, Bonds & Stocks

Nominal vs effective rates (EAR) · Loans: payments, amortization, balloon, DSCR · Terminal value and the double-counting trap · Bond valuation · Stock valuation: dividend models · EAR calculator: don't be fooled by nominal rates · Simulating Bank A vs Bank B loans · Bond prices move opposite to interest rates · Gordon model: why r – g dominates

- EAR = (1 + r/m)^m – 1; compare offers by EAR.
- Loan payments are annuities; early ones are mostly interest.
- Maturity shorter than amortization = balloon and refinancing risk.
- DSCR = cash for debt service ÷ annual debt service; below 1× is dangerous.
- Terminal value = next CF ÷ (r – g); avoid double counting.
- Bond prices fall when rates rise; long bonds are more sensitive.
- Stocks: Gordon $P_0 = D_1 \div (r - g)$; the r – g gap drives value.

FORMULAS

```
EAR = (1 + r/m)m - 1; continuous: er - 1
```

```
Payment = P × i ÷ [1 - (1+i)-N]
```

```
TV10 = CF11 ÷ (r - g), lalu PV = TV10 ÷ (1 + r)10
```

$$B_0 = I[1 - (1+rd)-n]/rd + M(1+rd)-n$$

$$P_0 = D_1 \div (r - g)$$